

Project Title

Advanced CSS3 & Responsive Portfolio Website Architecture

Objective

The objective of this project is to transform a semantic HTML5 portfolio website into a visually attractive, fully responsive, and modern web application using advanced CSS3 properties, CSS Grid, Flexbox, responsive media queries, and dynamic theme customization. The project focuses on creating a professional user interface that adapts seamlessly across mobile, tablet, and desktop devices.

Tools Used

- HTML5
- CSS3
- JavaScript
- Visual Studio Code / Notepad
- Google Chrome Browser
- Lighthouse Responsive Testing Tool

Dataset Description

No dataset is used in this project. Static portfolio details such as profile information, projects, education, and contact information are displayed in the website.

Machine Learning Algorithms Used

No machine learning algorithms are used in this project because it is based on frontend web development and responsive UI design.

Key Features

- CSS Grid for two-dimensional layouts

- Flexbox for component alignment
- Mobile-first responsive design
- Dynamic Light/Dark Mode Theme
- Responsive Media Queries
- Professional Portfolio UI Design
- Smooth Hover Effects and Animations
- Responsive Navigation Bar
- Accessible Contact Form

Steps Performed

1. Designed the semantic HTML5 webpage structure.
2. Implemented CSS Grid for overall page layout.
3. Used Flexbox for navigation and component alignment.
4. Added responsive media queries for different screen sizes.
5. Created CSS variables for light and dark mode themes.
6. Added animations, hover effects, and modern styling.
7. Designed responsive portfolio sections and project cards.
8. Tested website responsiveness on mobile, tablet, and desktop devices.

Output

- Developed a fully responsive professional portfolio website.
- Implemented CSS Grid and Flexbox layouts successfully.
- Added dynamic light/dark mode functionality.
- Improved website responsiveness across all devices.
- Created a visually modern and user-friendly interface.

Sample HTML & CSS Code

```
<!DOCTYPE html>
<html lang="en">
<head>
<meta charset="UTF-8">
<meta name="viewport" content="width=device-width, initial-
scale=1.0">

<title>Advanced Responsive Portfolio</title>
```

<style>

```
:root{
  --bg-color:#0f172a;
  --text-color:white;
  --card-color:#1e293b;
  --highlight:#38bdf8;
}

.light-mode{
  --bg-color:#f1f5f9;
  --text-color:#0f172a;
  --card-color:white;
  --highlight:#2563eb;
}

*{
  margin:0;
  padding:0;
  box-sizing:border-box;
  font-family:Arial,sans-serif;
}

body{
  background:var(--bg-color);
  color:var(--text-color);
  transition:0.5s;
}

/* Header */

header{
  background:var(--card-color);
  padding:20px;
  display:flex;
  justify-content:space-between;
  align-items:center;
  flex-wrap:wrap;
}

header h1{
```

```

        color:var(--highlight);
    }

    nav{
        display:flex;
        gap:20px;
    }

    nav a{
        color:var(--text-color);
        text-decoration:none;
        font-weight:bold;
    }

    nav a:hover{
        color:var(--highlight);
    }

    /* Theme Button */

    .theme-btn{
        background:var(--highlight);
        border:none;
        padding:10px 15px;
        border-radius:8px;
        cursor:pointer;
        color:white;
    }

    /* Grid Layout */

    .container{
        display:grid;
        grid-template-columns:repeat(2,1fr);
        gap:20px;
        padding:30px;
    }

    .card{
        background:var(--card-color);
        padding:25px;
        border-radius:15px;
    }

```

```
        box-shadow:0 5px 15px rgba(0,0,0,0.3);
        transition:0.3s;
    }

    .card:hover{
        transform:translateY(-5px);
    }

    .card h2{
        margin-bottom:15px;
        color:var(--highlight);
    }

    /* Contact Form */

    form{
        display:flex;
        flex-direction:column;
    }

    input,textarea{
        margin-top:10px;
        padding:12px;
        border:none;
        border-radius:8px;
    }

    button{
        margin-top:15px;
        padding:12px;
        border:none;
        border-radius:8px;
        background:var(--highlight);
        color:white;
        cursor:pointer;
    }

    /* Footer */

    footer{
        text-align:center;
        padding:20px;
```

```

        background:var(--card-color);
        margin-top:20px;
    }

    /* Mobile Responsive */

    @media(max-width:768px){

        .container{
            grid-template-columns:1fr;
        }

        header{
            flex-direction:column;
            text-align:center;
        }

        nav{
            margin-top:10px;
            flex-wrap:wrap;
            justify-content:center;
        }
    }

</style>
</head>

<body>

<header>
    <h1>My Portfolio</h1>

    <nav>
        <a href="#">Home</a>
        <a href="#">About</a>
        <a href="#">Projects</a>
        <a href="#">Contact</a>
    </nav>

    <button class="theme-btn" onclick="toggleTheme()">
        Light / Dark
    </button>

```

```
</header>
```

```
<div class="container">
```

```
    <section class="card">
      <h2>About Me</h2>
      <p>
        I am a frontend developer specialized in responsive web
design,
        CSS Grid, Flexbox, and accessibility standards.
      </p>
    </section>
```

```
    <section class="card">
      <h2>Skills</h2>
      <p>HTML5, CSS3, JavaScript, Responsive Design, UI/UX</p>
    </section>
```

```
    <section class="card">
      <h2>Projects</h2>
      <p>
        Developed responsive and modern portfolio websites with
advanced CSS3 layouts.
      </p>
    </section>
```

```
    <section class="card">
      <h2>Contact Form</h2>

      <form>

        <input type="text" placeholder="Enter Name">

        <input type="email" placeholder="Enter Email">

        <textarea rows="5" placeholder="Enter
Message"></textarea>

        <button type="submit">Send Message</button>

      </form>
    </section>
```

```
</div>

<footer>
  <p>© 2026 Advanced Responsive Portfolio Website</p>
</footer>

<script>

function toggleTheme(){
  document.body.classList.toggle("light-mode");
}

</script>

</body>
</html>
```

Conclusion

This project successfully demonstrates the implementation of advanced CSS3 styling techniques, responsive web design principles, and modern layout architectures using CSS Grid and Flexbox. The portfolio website adapts efficiently across different devices and provides an attractive, user-friendly, and professional interface.

Expected Outcome

- Fully responsive portfolio website
- Professional UI/UX design
- Dynamic light/dark mode support
- Mobile-first responsive architecture

OUTPUT

